

Below is a comprehensive classification of **diagram categories used in software development**, from requirements and architecture through implementation, testing, deployment, and operations.

1. Requirements & Business Analysis Diagrams

#	Diagram	Main Purpose
1	Use Case Diagram	Show actors and system functions
2	User Story Map	Organize user stories by user journey
3	User Journey Map	Visualize a user's experience
4	Customer Journey Map	Map customer interactions with a product
5	Context Diagram	Show system boundaries and external entities
6	Business Process Diagram	Model business processes
7	Business Model Diagram	Represent business structure/value
8	Capability Map	Show organizational/business capabilities
9	Mind Map	Explore requirements and ideas
10	Feature Map	Organize product features
11	Story Map	Connect activities, tasks, and stories

2. UML Diagrams

UML is one of the largest diagram families in software engineering.

A. UML Structural Diagrams

#	Diagram	Purpose
12	Class Diagram	Classes, attributes, methods, relationships
13	Object Diagram	Objects and their relationships
14	Component Diagram	Software components and dependencies
15	Composite Structure Diagram	Internal structure of classifiers
16	Package Diagram	Organize model elements into packages
17	Deployment Diagram	Map software to hardware/runtime nodes
18	Profile Diagram	Extend/customize UML
19	Structure Diagram	General structural representation

B. UML Behavioral Diagrams

#	Diagram	Purpose
20	Activity Diagram	Workflow and business/process logic
21	State Machine Diagram	States and transitions
22	Use Case Diagram	User/system interactions
23	Interaction Diagram	Communication between objects

C. UML Interaction Diagrams

#	Diagram	Purpose
24	Sequence Diagram	Time-ordered object interactions

25	Communication Diagram	Object communication
26	Interaction Overview Diagram	High-level interaction flow
27	Timing Diagram	State/value changes over time

3. Architecture Diagrams

#	Diagram	Purpose
28	System Architecture Diagram	Overall system structure
29	Application Architecture Diagram	Application-level architecture
30	Software Architecture Diagram	Major software building blocks
31	Solution Architecture Diagram	Complete solution structure
32	Enterprise Architecture Diagram	Organization-wide architecture
33	Logical Architecture Diagram	Logical system organization
34	Physical Architecture Diagram	Physical infrastructure
35	Conceptual Architecture Diagram	High-level architectural concepts
36	Technical Architecture Diagram	Technology implementation
37	Reference Architecture Diagram	Reusable architectural model
38	Integration Architecture Diagram	System integration
39	Security Architecture Diagram	Security boundaries and controls
40	Data Architecture Diagram	Data organization and movement
41	Network Architecture Diagram	Network topology
42	Cloud Architecture Diagram	Cloud services and infrastructure

4. C4 Model Diagrams

The C4 model provides multiple levels of architectural abstraction.

#	Diagram	Purpose
43	System Context Diagram	System and external users/systems
44	Container Diagram	Applications/services/databases
45	Component Diagram	Components inside a container
46	Code Diagram	Classes/code-level structure
47	Dynamic Diagram	Runtime interaction scenarios
48	Deployment Diagram	Deployment environment

5. Process & Workflow Diagrams

#	Diagram	Purpose
49	Flowchart	General process flow
50	Process Flow Diagram	Process steps
51	Workflow Diagram	Workflow execution
52	Swimlane Diagram	Process responsibilities

53	Cross-Functional Flowchart	Processes across departments/roles
54	Decision Tree	Decisions and possible outcomes
55	Decision Diagram	Decision logic
56	Activity Flow Diagram	Activities and transitions
57	State Flow Diagram	State transitions
58	Approval Workflow Diagram	Approval processes
59	BPMN Diagram	Business process modeling
60	Event-Driven Process Diagram	Event-based workflows

6. Data & Database Diagrams

#	Diagram	Purpose
61	ER Diagram (ERD)	Entities and relationships
62	Enhanced ER Diagram	Extended entity modeling
63	Database Schema Diagram	Tables and relationships
64	Relational Model Diagram	Relational database structure
65	Data Model Diagram	Data structures
66	Logical Data Model	Logical data relationships
67	Physical Data Model	Database implementation
68	Data Flow Diagram (DFD)	Data movement
69	Data Lineage Diagram	Data origin and transformation
70	Data Dependency Diagram	Data dependencies
71	Data Pipeline Diagram	Data processing pipeline
72	Data Warehouse Diagram	Warehouse architecture
73	Star Schema Diagram	Fact/dimension structure
74	Snowflake Schema Diagram	Normalized dimensional model

7. System & Data Flow Diagrams

#	Diagram	Purpose
75	System Context Diagram	System boundaries
76	Data Flow Diagram — Level 0	High-level data flow
77	DFD Level 1	Decompose major processes
78	DFD Level 2	Detailed process flow
79	Information Flow Diagram	Information movement
80	Control Flow Diagram	Control/program flow
81	Process Flow Diagram	Processing sequence
82	Input-Output Diagram	Inputs and outputs

8. API & Integration Diagrams

#	Diagram	Purpose
83	API Architecture Diagram	API ecosystem
84	API Flow Diagram	API request/response flow
85	REST API Diagram	REST interactions
86	GraphQL Architecture Diagram	GraphQL architecture
87	Integration Diagram	System integrations
88	Service Interaction Diagram	Service-to-service communication
89	Message Flow Diagram	Messaging
90	Event Flow Diagram	Event propagation
91	Sequence Diagram	API interaction sequence
92	Webhook Flow Diagram	Webhook processing
93	ESB Diagram	Enterprise service bus integration

9. Microservices Diagrams

#	Diagram	Purpose
94	Microservices Architecture Diagram	Services and relationships
95	Service Dependency Diagram	Service dependencies
96	Service Interaction Diagram	Service communication
97	Event-Driven Architecture Diagram	Events and consumers
98	Saga Pattern Diagram	Distributed transactions
99	CQRS Diagram	Command/query separation
100	Service Mesh Diagram	Service-to-service infrastructure
101	Distributed System Diagram	Distributed components
102	Circuit Breaker Flow Diagram	Failure handling
103	API Gateway Diagram	Gateway architecture

10. Object-Oriented Design Diagrams

#	Diagram	Purpose
104	Class Diagram	Classes and relationships
105	Object Diagram	Runtime objects
106	Inheritance Diagram	Class hierarchy
107	Interface Diagram	Interfaces and implementations
108	Dependency Diagram	Dependencies
109	Composition Diagram	Object composition
110	Aggregation Diagram	Aggregation relationships
111	Package Diagram	Package organization
112	Design Pattern Diagram	Design pattern structure

11. Software Design Pattern Diagrams

#	Diagram	Example
113	Singleton Diagram	One-instance object
114	Factory Diagram	Object creation
115	Abstract Factory Diagram	Object families
116	Builder Diagram	Complex object construction
117	Prototype Diagram	Object cloning
118	Adapter Diagram	Interface conversion
119	Decorator Diagram	Dynamic behavior extension
120	Facade Diagram	Simplified interface
121	Proxy Diagram	Controlled access
122	Observer Diagram	Publish/subscribe relationship
123	Strategy Diagram	Interchangeable algorithms
124	Command Diagram	Encapsulated operations
125	State Pattern Diagram	Behavior by state
126	Template Method Diagram	Algorithm skeleton

12. DevOps & CI/CD Diagrams

#	Diagram	Purpose
127	CI/CD Pipeline Diagram	Build-test-deploy process
128	DevOps Workflow Diagram	Development-to-operations flow
129	Build Pipeline Diagram	Build stages
130	Deployment Pipeline Diagram	Deployment stages
131	Release Flow Diagram	Release process
132	Git Branching Diagram	Branch strategy
133	Git Flow Diagram	Git-based development
134	Infrastructure Pipeline Diagram	Infrastructure automation
135	GitOps Diagram	Git-driven operations

13. Cloud & Infrastructure Diagrams

#	Diagram	Purpose
136	Cloud Architecture Diagram	Cloud resources
137	Infrastructure Diagram	Infrastructure components
138	Network Topology Diagram	Network connections
139	Deployment Diagram	Runtime deployment
140	Kubernetes Architecture Diagram	Kubernetes components
141	Container Architecture Diagram	Containers and services
142	Server Architecture Diagram	Servers and dependencies
143	Load Balancing Diagram	Traffic distribution
144	High Availability Diagram	Redundancy
145	Disaster Recovery Diagram	Recovery architecture

146	Multi-Region Architecture Diagram	Geographic deployment
147	Multi-Cloud Diagram	Multiple cloud providers

14. Security Diagrams

#	Diagram	Purpose
148	Security Architecture Diagram	Overall security design
149	Threat Model Diagram	Security threats
150	Attack Tree	Attack paths
151	Trust Boundary Diagram	Security boundaries
152	Data Flow Security Diagram	Secure data movement
153	Authentication Flow Diagram	Login/authentication
154	Authorization Flow Diagram	Access control
155	IAM Architecture Diagram	Identity/access management
156	Zero Trust Architecture Diagram	Zero-trust security
157	Network Security Diagram	Firewalls/security zones

15. Testing Diagrams

#	Diagram	Purpose
158	Test Flow Diagram	Testing workflow
159	Test Case Flowchart	Test execution
160	Test Strategy Diagram	Testing approach
161	Test Automation Architecture	Automation framework
162	Test Pyramid	Unit/integration/UI tests
163	Traceability Matrix	Requirements ↔ tests
164	State Transition Diagram	State-based testing
165	Decision Table Diagram	Decision-based testing
166	Cause-Effect Graph	Input/output relationships

16. Project Management Diagrams

#	Diagram	Purpose
167	Gantt Chart	Project schedule
168	PERT Chart	Project activity dependencies
169	Critical Path Diagram	Critical project activities
170	Dependency Graph	Task dependencies
171	Work Breakdown Structure	Decompose project work
172	Roadmap Diagram	Product/project timeline
173	Kanban Board	Work status
174	Burndown Chart	Remaining work
175	Burnup Chart	Completed work

176	Velocity Chart	Team delivery rate
177	Release Train Diagram	Release planning

17. Version Control Diagrams

#	Diagram	Purpose
178	Git Branch Diagram	Branch relationships
179	Git Flow Diagram	Feature/release/hotfix flow
180	Commit Graph	Commit history
181	Merge Flow Diagram	Branch merging
182	Release Branch Diagram	Release management

18. UI/UX Diagrams

#	Diagram	Purpose
183	Wireframe	UI structure
184	Screen Flow Diagram	Navigation between screens
185	User Flow Diagram	User interaction path
186	Information Architecture Diagram	Content organization
187	Site Map	Website structure
188	Navigation Map	Navigation relationships
189	Customer Journey Diagram	User experience
190	Interaction Flow Diagram	UI interactions
191	Prototype Flow	Prototype navigation

19. Algorithm & Programming Diagrams

#	Diagram	Purpose
192	Algorithm Flowchart	Algorithm steps
193	Pseudocode Flow Diagram	Algorithm logic
194	Control Flow Graph (CFG)	Program execution paths
195	Call Graph	Method/function calls
196	Dependency Graph	Code dependencies
197	Decision Tree	Conditional logic
198	State Machine	Program states
199	Recursion Diagram	Recursive execution
200	AST Diagram	Abstract syntax tree
201	Expression Tree	Expression structure

20. Runtime & Distributed-System Diagrams

#	Diagram	Purpose
202	Sequence Diagram	Runtime interactions
203	Timing Diagram	Timing relationships
204	Message Sequence Chart	Message ordering
205	Distributed System Diagram	Distributed components
206	Consensus Diagram	Distributed consensus
207	Replication Diagram	Data replication
208	Leader-Follower Diagram	Distributed roles
209	Event Sourcing Diagram	Event-based state
210	Publish-Subscribe Diagram	Event distribution
211	Queue Architecture Diagram	Message queues

21. Monitoring & Observability Diagrams

#	Diagram	Purpose
212	Observability Architecture Diagram	Logs, metrics, traces
213	Monitoring Architecture Diagram	Monitoring components
214	Logging Architecture Diagram	Log collection
215	Tracing Diagram	Distributed tracing
216	Metrics Flow Diagram	Metrics pipeline
217	Alert Flow Diagram	Alert generation
218	Incident Response Flowchart	Incident handling

22. Business Process Management

#	Diagram	Purpose
219	BPMN Process Diagram	Business processes
220	BPMN Collaboration Diagram	Participant interaction
221	BPMN Choreography Diagram	Message interactions
222	Value Stream Map	Value delivery
223	SIPOC Diagram	Supplier-input-process-output-customer
224	Process Hierarchy Diagram	Process decomposition

23. Enterprise Architecture Framework Diagrams

#	Diagram	Purpose
225	ArchiMate Motivation Diagram	Drivers/goals
226	ArchiMate Strategy Diagram	Strategy
227	ArchiMate Business Layer Diagram	Business architecture
228	ArchiMate Application Layer	Application architecture

	Diagram	
229	ArchiMate Technology Layer Diagram	Technology architecture
230	ArchiMate Physical Layer Diagram	Physical infrastructure
231	ArchiMate Implementation Diagram	Implementation/migration

24. Mathematical / Modeling Diagrams

#	Diagram	Purpose
232	State Transition Diagram	State changes
233	Decision Table	Rules
234	Truth Table	Boolean logic
235	Dependency Graph	Relationships
236	Directed Graph	Directed relationships
237	Undirected Graph	Undirected relationships
238	Tree Diagram	Hierarchical structure
239	DAG	Directed acyclic dependencies
240	Petri Net	Concurrent systems
241	Finite State Machine	Finite states
242	Sequence/Interaction Model	Event sequences

25. Documentation & Knowledge Diagrams

#	Diagram	Purpose
243	Concept Map	Connect concepts
244	Mind Map	Organize ideas
245	Knowledge Graph	Represent entities/relationships
246	Taxonomy Diagram	Classification
247	Hierarchy Diagram	Parent-child relationships
248	Glossary Relationship Diagram	Domain terminology
249	System Landscape Diagram	Systems across an organization

26. Migration & Transformation Diagrams

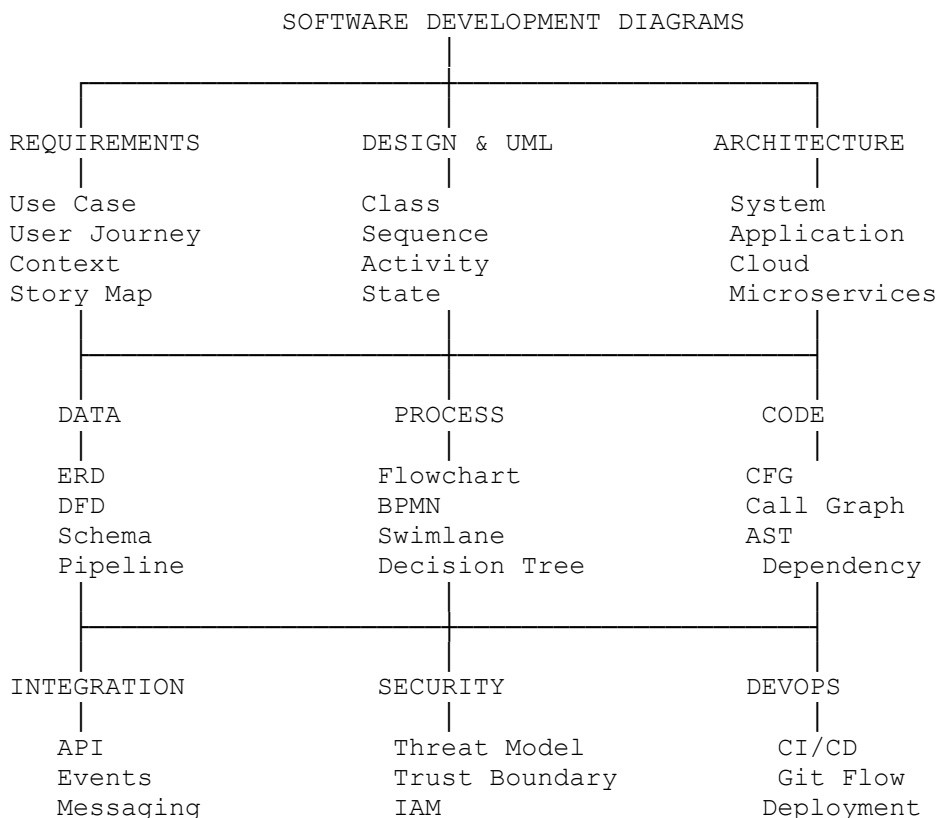
#	Diagram	Purpose
250	Migration Architecture Diagram	Migration design
251	As-Is Architecture Diagram	Existing system
252	To-Be Architecture Diagram	Target system
253	Transition Architecture Diagram	Migration stages
254	Data Migration Diagram	Data migration
255	Application Migration Diagram	Application migration
256	Modernization Roadmap	Legacy modernization

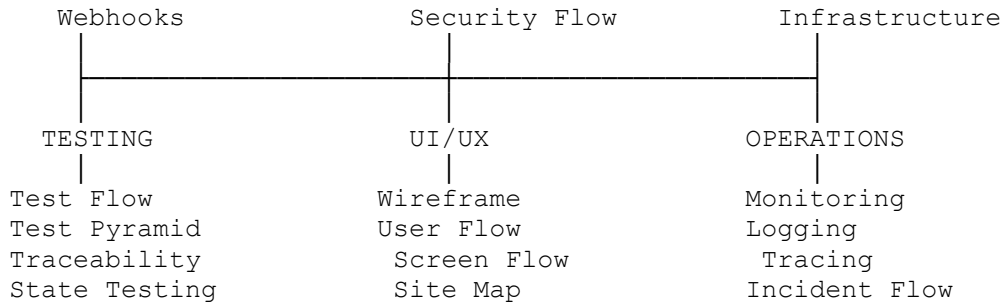
27. Specialized Software Diagrams

#	Diagram	Purpose
257	Domain Model Diagram	Domain concepts
258	Domain-Driven Design Context Map	Bounded contexts
259	Bounded Context Diagram	Domain boundaries
260	Event Storming Diagram	Domain events
261	Hexagonal Architecture Diagram	Ports and adapters
262	Clean Architecture Diagram	Architectural layers
263	Onion Architecture Diagram	Dependency direction
264	Layered Architecture Diagram	Application layers
265	MVC Architecture Diagram	Model-view-controller
266	MVVM Diagram	Model-view-viewmodel
267	MVP Diagram	Model-view-presenter
268	Serverless Architecture Diagram	Serverless components

A useful high-level classification

You can remember almost all software-development diagrams using these **12 major categories**:





Most important diagrams to learn first

If your goal is **software engineering / Java / backend development**, I would prioritize:

1. **Flowchart**
2. **Use Case Diagram**
3. **Class Diagram**
4. **Sequence Diagram**
5. **Activity Diagram**
6. **State Machine Diagram**
7. **Component Diagram**
8. **Deployment Diagram**
9. **ER Diagram**
10. **Data Flow Diagram**
11. **System Architecture Diagram**
12. **C4 Context Diagram**
13. **C4 Container Diagram**
14. **C4 Component Diagram**
15. **API Architecture Diagram**
16. **Microservices Architecture Diagram**
17. **CI/CD Pipeline Diagram**
18. **Cloud Architecture Diagram**
19. **Security Architecture Diagram**
20. **Domain/Context Map**

These cover the majority of diagrams you'll encounter in **requirements → analysis → design → coding → testing → deployment → operations**.

Yes. There are **many** tools, and no single product is equally strong for every diagram category. Below is a practical master list, grouped by what they are best at.

1. General-Purpose Diagramming Tools

These can handle flowcharts, architecture, UML, ERD, BPMN, network diagrams, process diagrams, etc.

Tool / Software	Free	Paid	Best for
diagrams.net (draw.io)	✓	—	General diagrams, UML, architecture, ERD
Lucidchart	Limited	✓	Business + technical diagrams
Microsoft Visio	—	✓	Enterprise/business diagrams
Miro	Limited	✓	Collaborative diagrams/whiteboarding
Creately	Limited	✓	Business + technical diagrams
Gliffy	Limited	✓	Confluence/Jira diagrams
EdrawMax	Limited	✓	Large diagram library
SmartDraw	—	✓	Business/engineering diagrams
Visual Paradigm	Community	✓	UML, BPMN, architecture
OmniGraffle	—	✓	macOS/iOS diagramming
ConceptDraw DIAGRAM	Trial	✓	Business/technical diagrams
yEd Graph Editor	✓	—	Graphs, networks, UML
Apache OpenOffice Draw	✓	—	Basic diagrams
LibreOffice Draw	✓	—	Basic diagrams
Dia	✓	—	Traditional technical diagrams
Pencil Project	✓	—	Wireframes/UI
Excalidraw	✓	Optional	Hand-drawn-style diagrams
Figma FigJam	Limited	✓	Collaborative diagrams/workshops

2. UML Diagram Tools

For Class, Object, Sequence, Activity, State, Component, Deployment, Package, etc.

Tool	Free	Paid
diagrams.net	✓	—
PlantUML	✓	—
Mermaid	✓	—
Visual Paradigm	✓Community	✓
StarUML	Trial	✓
Modelio	✓	✓
Eclipse Papyrus	✓	—
UMlet	✓	—
GenMyModel	Limited	✓
Astah	Community	✓
Enterprise Architect	—	✓

Sparx Systems Enterprise Architect	—	✓
IBM Rational Software Architect	—	✓
MagicDraw / Cameo Systems Modeler	—	✓
Lucidchart	Limited	✓
Visual Paradigm Online	Limited	✓

Particularly useful for Java developers

PlantUML + IntelliJ IDEA/VS Code/Eclipse is excellent for text-based UML.

Example:

@startuml

```
class Customer {
    -id : Long
    -name : String
    +placeOrder()
}
```

```
class Order {
    -orderId : Long
    -amount : double
}
```

```
Customer "1" --> "*" Order
```

@enduml

3. Mermaid Diagram Tools

Mermaid is particularly useful when diagrams should live alongside source code and documentation.

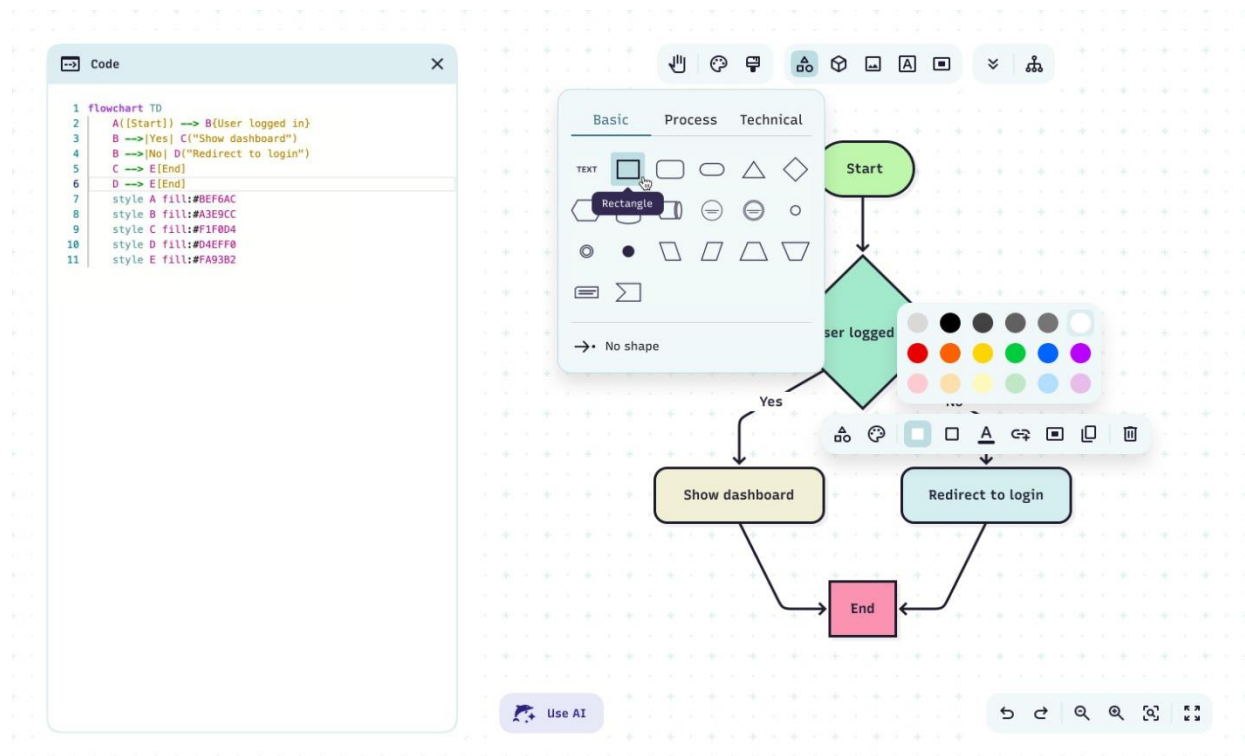
It supports:

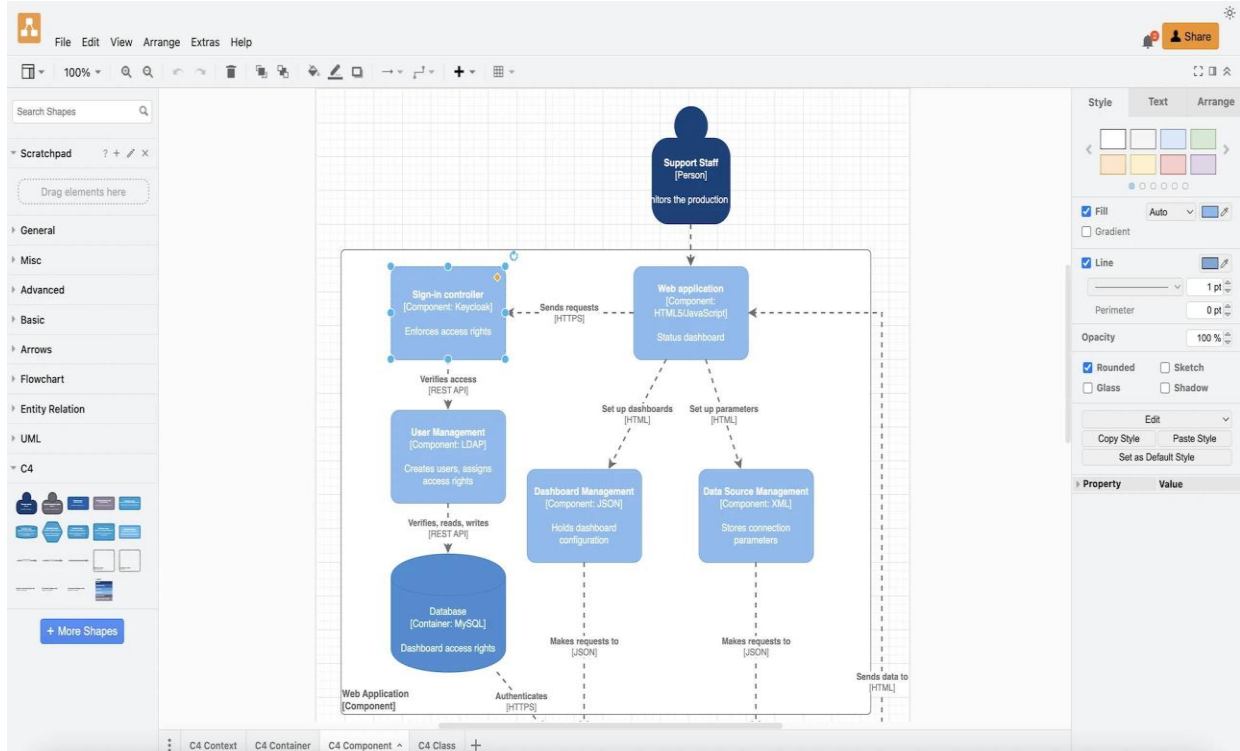
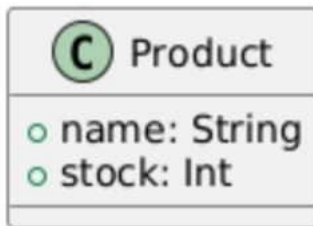
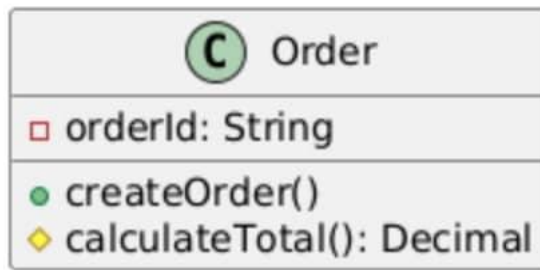
- Flowcharts
- Sequence diagrams
- Class diagrams
- State diagrams
- ER diagrams
- User journeys
- Gantt charts
- Git graphs
- Mindmaps
- Architecture diagrams
- Block diagrams

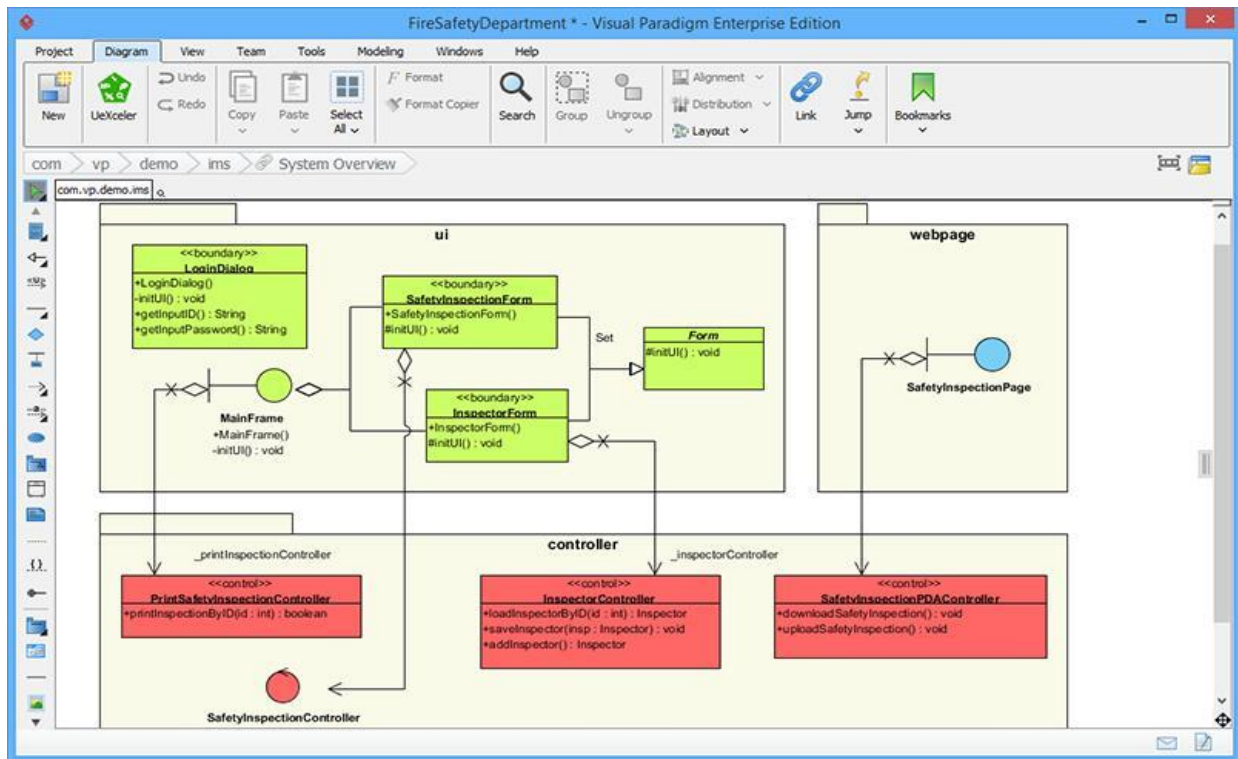
- Timeline diagrams
- Quadrant charts

Tools that support Mermaid

Tool	Free/Paid
Mermaid Live Editor	✓Free
GitHub	✓
GitLab	✓
VS Code + extensions	✓
Obsidian	✓
Notion	Depending on integration/workflow
JetBrains IDEs + plugins	✓/plugin-dependent
Docusaurus	✓
MkDocs + plugins	✓
Typora	Paid
Mermaid Chart	Free + Paid







4. PlantUML Tools

PlantUML is one of the most useful **code-based diagramming systems**.

It can create:

- Class diagrams
- Sequence diagrams
- Activity diagrams
- Component diagrams
- Deployment diagrams
- State diagrams
- Object diagrams
- Use case diagrams
- ER diagrams
- Timing diagrams
- ArchiMate diagrams
- Gantt charts
- Mind maps
- JSON/YAML diagrams

PlantUML ecosystem

Tool	Type
PlantUML	Free/Open Source
PlantUML Server	Free/Open Source
VS Code + PlantUML extension	Free
IntelliJ IDEA + PlantUML plugin	Free/plugin
Eclipse + PlantUML	Free/plugin
Visual Studio Code	Free
Kroki	Free/Open Source
PlantText	Free web editor
PlantUML QEditor	Free

5. Architecture Diagram Tools

For system architecture, application architecture, cloud, microservices, distributed systems, C4, infrastructure, etc.

Tool	Free	Paid
diagrams.net	✓	—
Lucidchart	Limited	✓
Miro	Limited	✓
Structurizr	Limited/Open	✓
IcePanel	Limited	✓
Eraser	Limited	✓
Mermaid	✓	—
PlantUML	✓	—
Archi	✓	—
Visual Paradigm	Community	✓
LeanIX	—	✓
Ardoq	—	✓
Enterprise Architect	—	✓

6. C4 Model Tools

For:

- System Context
- Container
- Component
- Code
- Dynamic
- Deployment

Recommended tools

Tool	Free	Paid
Structurizr	Limited	✓
PlantUML	✓	—
Mermaid	✓	—
IcePanel	Limited	✓
diagrams.net	✓	—
Eraser	Limited	✓
Visual Paradigm	Community	✓
Kroki	✓	—

Best choices: Structurizr, PlantUML, Mermaid, IcePanel.

7. ERD / Database Diagram Tools

For database design:

- Entity Relationship Diagrams
- Tables
- Primary keys
- Foreign keys
- Relationships
- Database schemas
- Normalization

Tool	Free	Paid
dbdiagram.io	Limited	✓
DBeaver	✓Community	✓
MySQL Workbench	✓	—
pgAdmin	✓	—
DataGrip	Trial	✓
SQL Server Management Studio	✓	—
Azure Data Studio / related Microsoft tooling	Depends	Depends
Oracle SQL Developer	✓	—
Oracle SQL Developer Data Modeler	✓	—
diagrams.net	✓	—
Lucidchart	Limited	✓
Visual Paradigm	Community	✓
Navicat Data Modeler	Trial	✓
ERwin Data Modeler	—	✓
SAP PowerDesigner	—	✓
Hackolade	Trial	✓

8. Data Flow Diagram Tools

For DFD Level 0, Level 1, Level 2, data movement and information systems.

Tool	Free	Paid
diagrams.net	✓	—
Lucidchart	Limited	✓
Visual Paradigm	Community	✓
Creately	Limited	✓
EdrawMax	Limited	✓
SmartDraw	—	✓
Microsoft Visio	—	✓
Mermaid	✓	—
Miro	Limited	✓

9. Flowchart Tools

Tool	Free	Paid
diagrams.net	✓	—
Mermaid	✓	—
Lucidchart	Limited	✓
Miro	Limited	✓
Creately	Limited	✓
Excalidraw	✓	Optional
yEd	✓	—
Visio	—	✓
EdrawMax	Limited	✓
SmartDraw	—	✓
Google Drawings	✓	—
LibreOffice Draw	✓	—

10. BPMN Tools

For Business Process Model and Notation.

Tool	Free	Paid
Camunda Modeler	✓	Enterprise options
Bizagi Modeler	✓	Enterprise options
Signavio	—	✓
Visual Paradigm	Community	✓
ProcessMaker	Community/limited	✓
bpmn.io	✓	—
diagrams.net	✓	—
Lucidchart	Limited	✓

Creately	Limited	✓
IBM Blueworks Live	—	✓
Bonita	Community	✓

11. UI/UX Diagram & Prototyping Tools

For:

- Wireframes
- User flows
- Screen flows
- Prototypes
- Information architecture
- UX journeys

Tool	Free	Paid
Figma	✓	✓
FigJam	Limited	✓
Penpot	✓Open Source	Paid cloud options
Sketch	—	✓
Adobe XD	—/availability varies	—
Balsamiq	Trial	✓
Axure RP	Trial	✓
Framer	Limited	✓
Justinmind	Limited	✓
MockFlow	Limited	✓
Pencil Project	✓	—
InVision	Availability varies	—
Miro	Limited	✓

12. Network Diagram Tools

For:

- Network topology
- Servers
- Routers
- Switches
- Firewalls
- VLANs
- Network architecture

Tool	Free	Paid
diagrams.net	✓	—
Microsoft Visio	—	✓
Lucidchart	Limited	✓
yEd	✓	—
EdrawMax	Limited	✓
SolarWinds Network Topology Mapper	—	✓
LanFlow	Trial	✓
ConceptDraw DIAGRAM	Trial	✓
NetBox	✓	—
Hava	Limited	✓

13. Cloud Architecture Diagram Tools

AWS

Tool	Free/Paid
AWS Architecture Icons	✓
AWS Application Composer	✓
AWS Infrastructure Composer	✓
diagrams.net	✓
Lucidchart	Limited/✓
Cloudcraft	Limited + Paid
Hava	Limited + Paid
Eraser	Limited + Paid

Microsoft Azure

Tool	Free/Paid
Azure Architecture Center icons	✓
Microsoft Visio	Paid
diagrams.net	✓
Lucidchart	Limited + Paid
Hava	Limited + Paid

Google Cloud

Tool	Free/Paid
Google Cloud Architecture Center	Resources
diagrams.net	✓
Lucidchart	Limited + Paid
CloudSkew	Free/Paid options
Hava	Limited + Paid

14. Kubernetes / Container Diagrams

Tool	Free	Paid
Kubernetes official icons	✓	—
diagrams.net	✓	—
Mermaid	✓	—
PlantUML	✓	—
IcePanel	Limited	✓
Eraser	Limited	✓
Miro	Limited	✓
K8sGPT ecosystem tools	Varies	Varies
InfraMap	✓	—
Monokle	Community/free options	✓
Lens	Free/Open Core	Paid options

15. Git / Version-Control Diagrams

For:

- Git Flow
- Branching
- Commit history
- Merge strategies
- Release flow

Tool	Free	Paid
Git Graph	✓	—
GitKraken	Limited	✓
Sourcetree	✓	—
GitHub	✓	Paid plans
GitLab	✓	Paid plans
Mermaid	✓	—
PlantUML	✓	—
diagrams.net	✓	—

16. Project Management Diagrams

For:

- Gantt
- PERT
- Critical path
- Roadmaps

- Dependencies
- WBS

Tool	Free	Paid
ProjectLibre	✓	—
GanttProject	✓	—
Microsoft Project	—	✓
Jira	Limited	✓
Azure DevOps	Limited	✓
Monday.com	Limited	✓
ClickUp	Limited	✓
Asana	Limited	✓
Smartsheet	—/trial	✓
Miro	Limited	✓
Mermaid	✓	—

17. Security / Threat Modeling Tools

For:

- Threat modeling
- Attack trees
- Trust boundaries
- Security architecture
- Data-flow security

Tool	Free	Paid
OWASP Threat Dragon	✓	—
Microsoft Threat Modeling Tool	✓	—
IriusRisk	Limited	✓
ThreatModeler	—	✓
securiCAD	—	✓
diagrams.net	✓	—
Lucidchart	Limited	✓
Miro	Limited	✓

18. Algorithm / Code Visualization Tools

For:

- Algorithms
- Control-flow graphs
- Call graphs
- ASTs

- Data structures
- Execution visualization

Tool	Free	Paid
Python Tutor	✓	—
VisuAlgo	✓	—
Mermaid	✓	—
PlantUML	✓	—
Graphviz	✓	—
D3.js	✓	—
yEd	✓	—
IntelliJ IDEA	Community/Ultimate	✓
Eclipse	✓	—
Visual Studio	Community	Paid editions

19. Graph Visualization Tools

For dependency graphs, knowledge graphs, network graphs, call graphs, etc.

Tool	Free	Paid
Graphviz	✓	—
Gephi	✓	—
Cytoscape	✓	—
yEd	✓	—
Mermaid	✓	—
D3.js	✓	—
Neo4j Bloom	Limited	✓
Kumu	Limited	✓
diagrams.net	✓	—

20. Knowledge Graph / Ontology Diagram Tools

Tool	Free	Paid
Neo4j Bloom	Limited	✓
Neo4j Browser	✓	—/platform dependent
Protégé	✓	—
Gephi	✓	—
yEd	✓	—
Graphviz	✓	—
Cytoscape	✓	—
OntoGraf	✓	—

21. Enterprise Architecture Tools

For large organizations and enterprise systems:

Tool	Free	Paid
Archi	✓	—
ArchiMate tools	Various	Various
Visual Paradigm	Community	✓
Sparx Enterprise Architect	—	✓
BiZZdesign Horizzon	—	✓
LeanIX	—	✓
Ardoq	—	✓
MEGA HOPEX	—	✓
Orbus iServer	—	✓
Avolution ABACUS	—	✓
IBM Rational tools	—	✓
Cameo Enterprise Architecture	—	✓

22. DDD / Domain Modeling Tools

For Domain-Driven Design:

- Domain models
- Bounded contexts
- Context maps
- Aggregates
- Domain events
- Event storming

Tool	Free	Paid
Miro	Limited	✓
FigJam	Limited	✓
diagrams.net	✓	—
PlantUML	✓	—
Mermaid	✓	—
Structurizr	Limited	✓
IcePanel	Limited	✓
EventStorming boards	Various	Various

23. Whiteboard / Collaborative Diagram Tools

Tool	Free	Paid
Miro	Limited	✓
FigJam	Limited	✓

Microsoft Whiteboard	✓/Microsoft plans	✓
Excalidraw	✓	Optional
Lucidspark	Limited	✓
Creately	Limited	✓
Whimsical	Limited	✓
tldraw	✓	—

24. Documentation Tools with Diagram Support

Tool	Free	Paid
GitHub Markdown	✓	Paid plans
GitLab	✓	Paid plans
Notion	Free/limited	✓
Confluence	—/limited	✓
Obsidian	✓	Optional
Docusaurus	✓	—
MkDocs	✓	—
AsciiDoc	✓	—
BookStack	✓	—
Outline	Open Source/paid cloud	✓

These can integrate with **Mermaid**, **PlantUML**, **Graphviz**, **draw.io**, or other diagram technologies.

25. Text-to-Diagram / Diagram-as-Code Tools

This category is especially useful for developers.

Tool	Free	Paid
Mermaid	✓	Optional
PlantUML	✓	—
Graphviz	✓	—
D2	✓	—
Structurizr DSL	✓	✓
nomnoml	✓	—
Kroki	✓	—
ASCII diagrams	✓	—
Diagrams Python library	✓	—
Python Graphviz	✓	—

Example with Mermaid:

```

flowchart LR
    User --> API

```

API --> Service
Service --> Database

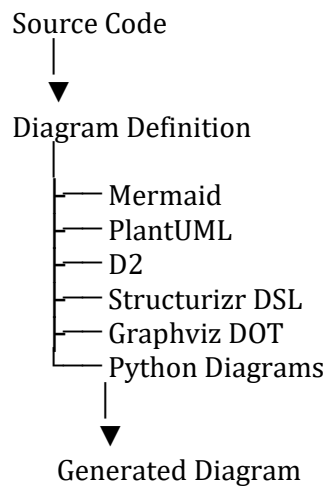
26. Infrastructure-as-Code Diagram Tools

Useful for visualizing infrastructure defined through code.

Tool	Free/Paid
Python Diagrams	✓
Terraform + diagram tools	Mostly free/open source
Inframap	✓
Blast Radius	✓
Terraform Graph	✓
Terravision	Open source
Hava	Limited + Paid
Cloudcraft	Limited + Paid
Brainboard	Paid/limited
env0	Paid/limited
Spacelift	Paid/limited

27. Architecture-as-Code / Diagram-as-Code

A growing category:



This approach is particularly valuable because diagrams can be:

- Version controlled
- Reviewed in Git
- Automatically generated
- Updated alongside code

- Included in CI/CD
- Embedded into documentation

28. Best Free Tools

If you want a **100% free software-development diagram toolkit**, I'd recommend:

Requirement	Tool
General diagrams	diagrams.net
UML	PlantUML
Text-based diagrams	Mermaid
Graphs	Graphviz
Graph visualization	yEd
Database	DBeaver / MySQL Workbench
BPMN	Camunda Modeler
UI/UX	Penpot / Figma free tier
Whiteboarding	Excalidraw
Cloud architecture	diagrams.net
AWS diagrams	diagrams.net + AWS icons
Azure diagrams	diagrams.net + Azure icons
GCP diagrams	diagrams.net + GCP icons
Threat modeling	OWASP Threat Dragon
Algorithm visualization	VisuAlgo / Python Tutor
Enterprise Architecture	Archi
Git diagrams	Sourcetree / Git Graph
Project Gantt	ProjectLibre / GanttProject
Kubernetes	diagrams.net / Mermaid
DDD	Miro free / diagrams.net

29. Best Paid Tools

For professional/enterprise environments:

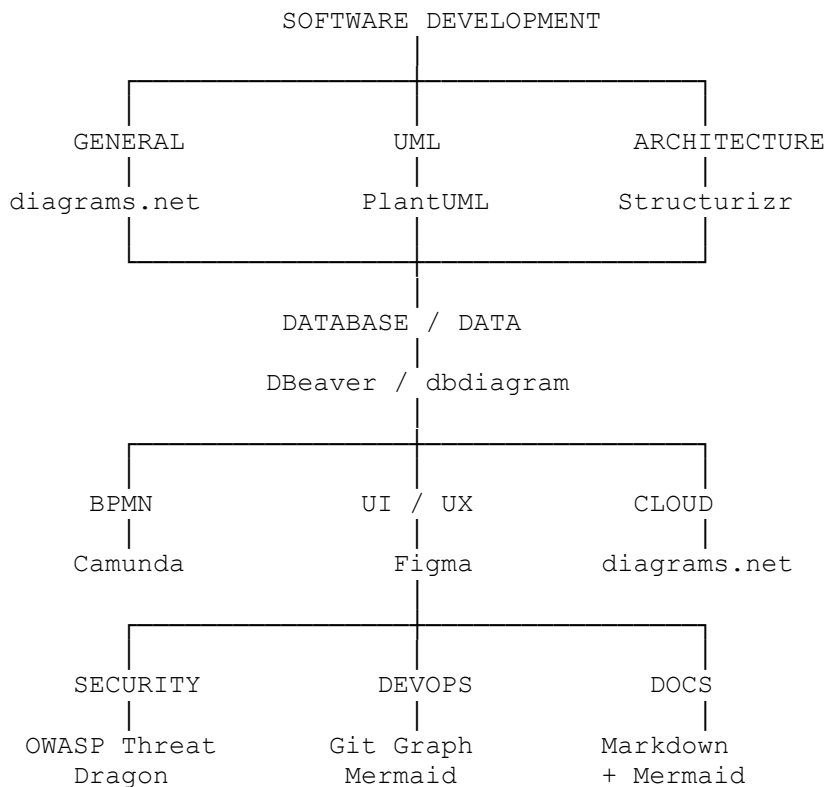
Need	Recommended
Enterprise diagramming	Microsoft Visio
Collaborative diagramming	Lucidchart
Enterprise architecture	LeanIX / Ardoq / HOPEX
UML	Visual Paradigm / Enterprise Architect
Advanced UML/SysML	Cameo Systems Modeler
BPMN	SAP Signavio / Camunda
UI/UX	Figma / Axure RP
Cloud architecture	Cloudcraft / Hava
Database modeling	ERwin / PowerDesigner
Team whiteboarding	Miro

Product/project planning	Jira / Microsoft Project / Monday.com
Threat modeling	IriusRisk / ThreatModeler

30. My Recommended Software Developer Stack

If you're learning **Java + Spring Boot + Microservices + SQL + Cloud**, you don't need 50 applications.

A very strong stack is:



If I had to choose only 7

1. **diagrams.net** — general-purpose diagrams
2. **PlantUML** — UML and software design
3. **Mermaid** — diagrams-as-code/documentation
4. **DBeaver** — database/ER diagrams
5. **Figma** — UI/UX
6. **Camunda Modeler** — BPMN/process modeling
7. **Archi** — enterprise architecture

That combination covers **most software-development diagram categories without requiring expensive software**.